Post-hoc Entropy Minimization Improves Semi-Supervised Medical Image Segmentation

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***Abstract*—Semi-supervised learning methods for medical image segmentation, or SSL methods, stop using the unlabeled data once training converges, even though the model's predictions on it still carry residual uncertainty concentrated at organ boundaries. This paper shows that this residual entropy is spatially structured and exploits it with *Post-hoc Entropy Minimization*, or PEM: any converged SSL checkpoint is fine-tuned for two to five epochs with a single entropy-minimization loss on unlabeled data, using no labels, no pseudo-labels, and no architectural changes. A label-free diagnostic ρ_f, computed before fine-tuning, separates the checkpoints PEM improves from the one it destroys, and a gradient analysis explains why the loss localizes to the boundary with no explicit mask, and why masking nonetheless matters, as insurance against collapse. In a corrected retrospective re-analysis, the learning rate and mask threshold are chosen on the labeled training volumes, which no post-hoc objective sees, rather than on test data. Applied to Bidirectional Copy-Paste, or BCP, checkpoints, released for Left Atrium and reproduced for Pancreas-CT, PEM reduces the 95th-percentile Hausdorff Distance by 25 to 44% and improves the Dice score by $+1.02$, $+0.66$, and $+0.62$ points on Pancreas-CT at 20% labeled, Left Atrium at 5% labeled, and Left Atrium at 10% labeled, in minutes per run on a single GPU. Against temperature scaling, pseudo-label fine-tuning, and self-consistency under an identical protocol and selection rule, the comparison separates by dataset: in exploratory retrospective tests on Pancreas-CT, PEM improves HD95 over BCP and both post-hoc baselines (Holm-adjusted $p \leq .003$), while on Left Atrium its HD95 gains over BCP do not survive correction and no significant difference from pseudo-label fine-tuning is detected on either metric. A mask ablation at a fixed budget identifies one threshold that degrades none of the SSL checkpoints, where looser ones cost up to 30 Dice points.**

***Index Terms*—semi-supervised learning, medical image segmentation, entropy minimization, post-hoc fine-tuning**

I. Introduction

Voxel-level annotation of 3D scans is slow and expensive, making semi-supervised learning, or SSL, from a small labeled and a large unlabeled set a practical necessity for volumetric segmentation. Recent SSL methods for this task [1]–[5] reach strong accuracy by using the unlabeled data during training and then discarding it at convergence.

This paper makes a complementary observation: at convergence a well-trained SSL model still assigns non-trivial prediction entropy to the unlabeled volumes, concentrated in a thin shell of low-confidence voxels on the predicted organ boundary. This concentration, measured here on BCP checkpoints, is exactly the condition under which a label-free sharpening step is well-behaved, the gradient it produces being already aligned with the locus of segmentation error. This is exploited with *Post-hoc Entropy Minimization*, or PEM: a converged checkpoint is fine-tuned for a few epochs on the unlabeled data with a single entropy objective, using no labels, pseudo-labels or architectural changes, in minutes on one GPU. Entropy minimization is classical [6] and standard in modern SSL [7]; what is new is its operating regime and the three results that follow.

A label-free diagnostic. The first contribution is the *shell entropy concentration* ρ_f, a scalar that turns the observation above into a test. For a converged checkpoint it is the fraction of total prediction entropy carried by the low-confidence voxels of the unlabeled set, so it needs neither labels nor a validation set and follows from a single forward pass. On the four checkpoints evaluated it separates those on which post-hoc sharpening is well-conditioned from the one on which it is not: the three BCP checkpoints hold 89 to 95% of their entropy in a shell of a few percent of the voxels, whereas a supervised-only network on the same labels keeps 55% there and scatters the rest through the interior (Table I).

PEM and why it works. The second contribution is PEM with an account of why the classical loss needs no stabilizers here. The entropy gradient is supported on a moderate-confidence band flanking the boundary and vanishes on the saturated interior and at maximum ambiguity, so it gates itself to the shell; the confidence mask further removes residual high-confidence gradients and renormalizes over the retained voxels, acting as insurance against collapse rather than a source of gain. Because partial-volume averaging and weak soft-tissue contrast make anatomical-boundary voxels the ambiguous ones, the decision surface co-locates with the organ surface, so entropy minimization realizes the cluster assumption as boundary refinement, and the argument predicts two regimes in which PEM should fail.

A corrected, retrospective evaluation. The third contribution is an empirical study over two organs and two modalities, reanalysed so that learning rates and thresholds are chosen on the labeled training volumes rather than on test data, for PEM and every baseline alike. PEM improves all four metrics on all three BCP checkpoints (Table II); exploratory,

retrospective per-case tests find the boundary gain significant against every competitor on Pancreas-CT but not on Left Atrium, where no significant difference between PEM and pseudo-label fine-tuning is detected (Table IV). The dataset-specific selected settings degrade none of the SSL checkpoints, and the Pancreas-selected setting destroys the one below the suitability threshold.

II. Related Work

SSL for medical image segmentation. Two mechanisms dominate: *consistency regularization*, where a student agrees with a teacher or itself under perturbations, and *pseudo-labeling*, where confident predictions become hard targets. Mean Teacher [1] uses an EMA teacher; UA-MT [2] filters by uncertainty; DTC [8] and MC-Net+ [9] enforce consistency across heads or decoders; CPS [10] combines both; BCP [3] mixes labeled and unlabeled crops bidirectionally, while MOST [4] and DyCON [5] improve further but release training code only. All share a training-time contract: the unlabeled data is used during optimization and never revisited. PEM revisits what it still contains at convergence.

Entropy minimization and test-time adaptation. Entropy minimization [6] implements low-density separation [11]; Pseudo-Label [12] is its hard-target limit and FixMatch [7] sharpens predictions into pseudo-targets, all *during* training and coupled to a loss preventing collapse. The closest methodology is test-time adaptation: TENT [13] minimizes entropy on a test stream updating only normalization parameters, SHOT [14] adds information maximization, and EATA [15] and SAR [16] add entropy-based selection, the masking question of Table III, documenting the collapse unrestricted minimization can cause. All assume distribution shift. PEM uses the same loss on a converged checkpoint and training-distribution data, where the residual entropy is already boundary-concentrated, so all parameters can be updated.

Post-hoc refinement. Temperature scaling [17] is monotone and cannot move an argmax, so it is included only to rule out recalibration; stochastic weight averaging [18] and self-distillation [19] need the training trajectory or hard targets, closer to the pseudo-label baseline.

III. Method

A. Shell Entropy Concentration

Let f_{θ^*} be a converged checkpoint and $\mathbf{x} \in \mathcal{U}$ an unlabeled volume. For voxel i with per-class softmax probabilities $p_{i,c}$, define the per-voxel entropy $H_i = -\sum_c p_{i,c} \log p_{i,c}$ and confidence $c_i = \max_c p_{i,c}$. The *shell entropy concentration* at confidence threshold τ is

$$\rho_f(\tau) = \frac{\mathbb{E}_{\mathbf{x}\sim\mathcal{U}}[\sum_i H_i \mathbb{1}[c_i \le \tau]]}{\mathbb{E}_{\mathbf{x}\sim\mathcal{U}}[\sum_i H_i]} \in [0,1]. \quad (1)$$

$\rho_f(\tau)$ is the fraction of total prediction entropy carried by voxels of confidence at most τ. It is label-free and scalar but carries no spatial information; that the low-confidence voxels lie near the boundary is checked separately, through $\bar{d}$ (Table I). It is always evaluated at $\tau = 0.99$, with $\rho^\star = 0.85$ a provisional suitability threshold placed below the SSL checkpoints and well above the supervised baseline of Table I. The loss below uses a *separate* threshold τ_m: τ measures where the entropy sits, τ_m decides which voxels the loss averages over.

B. Post-hoc Entropy Minimization

PEM fine-tunes the checkpoint on the unlabeled set with the entropy loss

$$\mathcal{L}_{\text{PEM}} = \frac{1}{|\mathcal{U}|} \sum_{\mathbf{x}\in\mathcal{U}} \frac{\sum_i m_i H_i}{\sum_i m_i + \epsilon}, \qquad m_i = \mathbb{1}[c_i \le \tau_m], \quad (2)$$

restricted to the confidence shell through m_i, with ϵ guarding the empty-mask case. No labels or pseudo-labels are used, and fine-tuning runs for a small fixed E at a small learning rate.

IV. Analysis: Why PEM Works

The gradient. The analysis is stated for a binary voxel with logit z and probability $p = \sigma(z)$; the multi-class case replaces $z\,p(1-p)$ with the analogous per-logit softmax term. The per-voxel entropy gradient is

$$\frac{\partial H(p(z))}{\partial z} = \frac{\partial H}{\partial p} \cdot \frac{\partial p}{\partial z} = (-z) \cdot p(1-p) = -\,z\,p(1-p). \quad (3)$$

Proposition 1 (Boundary-band gradient support): For the unmasked entropy loss, the magnitude of each voxel's gradient is proportional to $|z_i|\,p_i(1-p_i)$, which vanishes at $p_i \to 0$, at $p_i \to 1$, and at $p_i = 0.5$, and is maximized at $|z_i| \approx 1.54$. The gradient is thus automatically supported on the moderate-confidence band flanking the decision surface, with no explicit confidence mask.

What the mask does. Proposition 1 describes the unmasked loss. The mask changes it in two ways at once: it zeroes the gradient of voxels above τ_m, which just above 0.90 is still 88% of its maximum, and Eq. 2 renormalizes over the retained voxels instead of a volume 91 to 95% saturated, a 12 to $20\times$ dilution at $\tau = 0.99$. This ablation does not isolate the two. What the mask buys is asymmetric: on the instance-normalized Pancreas checkpoint the unmasked loss is better, by 0.12 and 0.08 Dice at the two learning rates, while on the batch-normalized LA checkpoints it costs up to 11 Dice points, and the loosest threshold 30. The mask is therefore insurance rather than a source of gain.

Two bounded properties. (i) With the gradient vanishing on committed voxels, a high-ρ_f checkpoint is stable at a small rate and tight mask; looser masks, larger rates or low ρ_f break this. *(ii)* The mean gradient scales with $\bar{V} = \mathbb{E}_i[p_i(1-p_i)]$, so as predictions commit and $\bar{V} \to 0$ the gain diminishes and pseudo-label fine-tuning, whose gradient does not vanish on committed voxels, becomes competitive.

Refinement, not relocation. The step has the sign of z_i, committing a boundary voxel to its more probable class, so it refines a boundary that SSL convergence placed correctly and reinforces the error otherwise. This is why the relative effect is largest on the surface metrics. ρ_f measures how much entropy the low-confidence voxels hold; that they form a boundary

shell rather than scattering through the interior is shown by $\bar{d}$ and the entropy maps.

V. Datasets and Implementation

Datasets. Pancreas-CT [20] (abdominal CT) is split into 12 labeled, 50 unlabeled and 18 test cases (20% labeled); Left Atrium (LA) 2018 (cardiac MRI) at 5% (4 labeled) and 10% (8 labeled) with 20 test cases. Both use the preprocessing distributed with BCP [3].

Checkpoints and protocol. Experiments start from the publicly released BCP checkpoints for LA (batch normalization) and, for Pancreas, from our reproduction with the official BCP recipe (instance normalization; 82.89 Dice against 82.91 reported), saved at its best-test-Dice evaluation, so its base number is optimistic; every post-hoc method starts from it. All are V-Nets. PEM uses Adam with gradient clipping at 1, no weight decay or schedule, and LA batch-normalization statistics frozen. The budgets $E = 2$ (Pancreas) and $E = 5$ (LA) are retained from the submitted version, where they were set with test-set learning curves in view; they are not re-selected, and the final epoch is reported. Table II and the supervised control are means over seeds $\{2020, 42, 123\}$; the grid of Table III uses seed 2020. Inference uses sliding windows with BCP's strides and no test-time augmentation; the fine-tuning itself takes under a minute on one RTX 4070 Ti SUPER, sliding-window evaluation dominating run time.

Selection protocol. No post-hoc method here optimizes on labels, so the labeled training volumes (12 Pancreas, 4 and 8 LA) are excluded from every post-hoc optimization objective but used to select the learning rate and mask threshold. Five-fold cross-validation is impossible with four volumes, so they serve as a non-independent selection set without cross-validation; because they score competing configurations and also trained the base checkpoint, selection overfitting remains possible. Selection maximizes the Dice *change* these volumes report, never their absolute Dice, which is optimistic because the base checkpoint was fitted on them. Checked retrospectively over 33 configurations, the 30 of Table III and three learning-rate-ceiling runs, that change ranks configurations as the test change does, with Spearman ρ of 0.78 (Pancreas), 0.75 (LA 5%) and 0.99 (LA 10%), and 0.67 to 0.96 once outright collapses are removed, so it is not mere collapse detection. The rule selects $\tau_m = 0.90$ in all three settings, at 5×10^{-5} (Pancreas) and 5×10^{-6} (LA), and every baseline is put through it over the identical grid.

Metrics and post-hoc baselines. Metrics are Dice and Jaccard (higher better) and HD95 and Average Surface Distance (ASD) in voxels (lower better). PL-FT and SC receive the same checkpoint, budget, data and selection rule, and TS serves as a recalibration check: *Temperature Scaling* (TS) fits a scalar temperature and cannot change the argmax; *Pseudo-Label Fine-Tuning* (PL-FT) generates hard pseudo-labels (argmax then largest connected component) and fine-tunes with cross-entropy plus Dice; *Self-Consistency* (SC) minimizes the squared error between the student on a noised input and the frozen teacher on the clean one. The comparison is against BCP because MOST [4] and DyCON [5] release training code only.

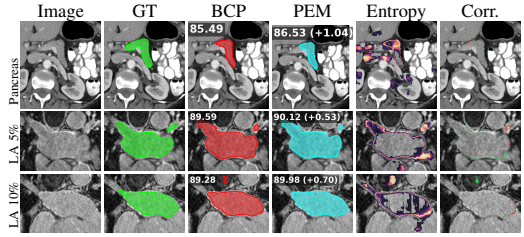


Fig. 1. The *median*-improvement test case per configuration (seed-2020 models), so no case is cherry-picked (per-case Dice overlaid; parenthesis = PEM gain). Columns: image, ground truth (green), BCP (red), BCP+PEM (cyan); **Entropy** of BCP (magma: the boundary shell); **Corr.** (green = PEM fixes, red = PEM breaks).

TABLE I
Shell entropy concentration on unlabeled training volumes. **Sat%**: voxels with $c_i > 0.99$. $\bar{d}$: mean distance from uncertain voxels to the predicted boundary. The SSL checkpoints clear $\rho^\star$=0.85; the supervised baseline does not.

Checkpoint	Sat%	Unc%	$\rho_f(0.99)$	$\bar{d}$
Supervised V-Net (no SSL)	90.7	9.3	0.552 ✗	14.45
BCP Pancreas-CT 20%	**95.0**	5.0	**0.894** ✓	10.56
BCP LA 5%	91.8	8.2	**0.948** ✓	5.19
BCP LA 10%	91.4	8.6	**0.945** ✓	5.96

VI. Experiments and Results

Residual entropy is concentrated (Table I). The three evaluated BCP checkpoints carry 89–95% of their prediction entropy in a shell of only 5–9% of the unlabeled voxels, giving $\rho_f(0.99) \in \{0.894, 0.948, 0.945\}$, all clearing $\rho^\star = 0.85$. A supervised-only V-Net on the same 12 Pancreas-CT labels carries 55% there ($\rho_f = 0.552$), its uncertain voxels lying 14.5 voxels from any boundary against 5–11, and does not clear it. Under the dataset-specific selected settings PEM improves all three BCP checkpoints; at the Pancreas-selected setting ($\tau_m = 0.90$, 5×10^{-5}, $E = 2$) it collapses this control from 75.66 to 17.29 ± 5.47 Dice (mean $\pm$ SD, three seeds), and the unmasked loss degrades it too. The diagnostic thus separates these four checkpoints under the selected protocol: with no shell, sharpening reinforces errors, mask or not.

PEM refines boundaries (Table II). TS moves Dice by less than 0.04, so none of this is recalibration and its row is omitted. PEM improves every metric on all three checkpoints, by +1.02, +0.66 and +0.62 Dice and by 24.8%, 44.0% and 30.5% of HD95, with seed SD of 0.06, 0.19 and 0.07. Its mean HD95 is also the best of the post-hoc methods on Pancreas and LA 5%, but only the Pancreas advantage is significant in the exploratory tests of Table IV. The metrics move on different timescales: on LA 5% (seed 2020) HD95 falls from 13.91 to 9.18 voxels in one epoch, over three quarters of its total gain, while Dice dips to 87.11 before recovering to 88.03 by epoch 5.

Exploratory per-case tests (Table IV). Because the budgets, grid bounds and Pancreas base checkpoint were informed by

TABLE II

POST-HOC METHODS ON THE THREE EVALUATED BCP CHECKPOINTS. EVERY METHOD GETS THE SAME CHECKPOINT, BUDGET, DATA AND LEARNING-RATE GRID, AND HAS ITS RATE CHOSEN BY THE LABELED-VOLUME SELECTION RULE OF SEC. V; THE TEST SPLIT IS NOT USED FOR THAT CHOICE. MEANS OVER SEEDS $\{2020, 42, 123\}$; BOLD MARKS THE BEST PER COLUMN. TS OMITTED ($|\Delta\text{DICE}|<0.04$). NO SIGNIFICANT PEM–PL-FT DIFFERENCE IS DETECTED ON LA. DIFFERENCES QUOTED IN THE TEXT USE UNROUNDED MEANS.

Method	Dice↑	Jaccard↑	HD95↓	ASD↓
Pancreas-CT 20% (12L / 50U / 18T)				
BCP (reproduced)	82.89	71.01	7.81	2.62
+ PL-FT	83.18	71.43	7.68	2.44
+ SC	83.04	71.23	7.62	2.49
+ **PEM**	**83.91**	**72.50**	**5.87**	**1.74**
LA 5% (4L / 76U / 20T)				
BCP (released ckpt)	87.32	77.65	13.91	3.60
+ PL-FT	**88.24**	**79.06**	10.11	2.68
+ SC	87.75	78.29	11.77	3.08
+ **PEM**	87.99	78.68	**7.79**	**2.28**
LA 10% (8L / 72U / 20T)				
BCP (released ckpt)	89.40	80.92	9.88	2.86
+ **PL-FT**	**90.09**	**82.05**	**6.37**	**1.84**
+ SC	89.58	81.24	8.04	2.30
+ PEM	90.02	81.95	6.87	2.19

test results, and Holm correction does not remove that post-selection bias, these tests are exploratory. Each case's metric is averaged over the three seeds before testing. On Pancreas PEM lowers HD95 against BCP on 17 of 18 cases and beats PL-FT and SC after correction ($p_H \leq .003$), while Dice improves more weakly ($p_H = .042$), the ordering a boundary-refinement step should produce. On LA the HD95 gains over BCP are nominal only ($p_H = .059$), Dice is significant on LA 10% alone, and no significant difference from PL-FT is detected on either metric; its mean HD95 lead on LA 5% rests on a few cases, PEM being better on only 9 of 20. PEM's Dice margin over PL-FT tracks headroom, $+0.73$, -0.26, -0.07 at base Dice 82.89, 87.32, 89.40, which is consequence (ii) of Sec. IV.

What the qualitative maps show (Fig. 1). The entropy column shows the per-voxel entropy H_i that ρ_f summarizes: a thin bright rim on the predicted surface over a dark interior. Corrections of either sign lie on the surface in all three settings, as the mechanism claims, but their balance tracks headroom: green outnumbers red on LA 5%, corrections are sparse but boundary-located on Pancreas, and on the saturated LA 10% red covers part of the rim even in a case PEM improves overall.

Mask and threshold ablation (Table III). Loosening τ_m from 0.90 to 0.999 readmits the interior and approaches the unmasked loss, and the count of degraded cells rises monotonically along that path, from 0 to 3, and failures reach 30 Dice points. $\tau_m = 0.95$ attains the largest single gain, but its advantage over $\tau_m = 0.99$ is unstable at a fixed budget, three cells won and three lost, by as little as 0.03. $\tau_m = 0.90$ is recommended: it alone never degrades a BCP checkpoint and has the highest mean over the six cells ($+0.72$ against $+0.46$). On LA, rates of 10^{-6} and below leave Dice unchanged, and at 5×10^{-5} the batch-normalized LA checkpoints lose 80 and 85 Dice while instance-normalized Pancreas is stable. On Pancreas a disagreement mask [2] is empty on a converged checkpoint; with the unmasked loss, decoder-only updates gain less than full fine-tuning ($+0.65$ against $+1.12$ Dice) and EMA updates almost nothing ($+0.06$), so PEM updates all student parameters.

TABLE III

MASK AND LEARNING-RATE ABLATION, AS ΔDICE AT EACH DATASET'S FIXED ENDPOINT (E=5 LA, E=2 PANCREAS) RATHER THAN AT A PEAK OVER THE BUDGET. **DEG.** COUNTS CELLS IN WHICH THE CHECKPOINT IS DEGRADED: ONLY τ_m=0.90 DEGRADES NONE, AND FRAGILITY GROWS MONOTONICALLY AS THE MASK LOOSENS.

Mask	LA 5%		LA 10%		Pancreas 20%		Deg.
	$5{\cdot}10^{-6}$	10^{-5}	$5{\cdot}10^{-6}$	10^{-5}	10^{-5}	$5{\cdot}10^{-5}$	
Full (no mask)	$+0.66$	-0.53	-0.86	-11.3	$+0.53$	$+1.12$	3
Confident τ_m=0.90	$+0.70$	$+1.20$	$+0.63$	$+0.36$	$+0.41$	$+1.04$	**0**
Confident τ_m=0.95	$+0.87$	$+1.43$	$+0.55$	-1.58	$+0.44$	$+1.04$	1
Confident τ_m=0.99	$+0.94$	$+0.40$	-1.17	-16.8	$+0.50$	$+1.07$	2
Confident τ_m=0.999	$+0.74$	-0.81	-2.57	-30.2	$+0.53$	$+1.11$	3

TABLE IV

EXPLORATORY RETROSPECTIVE ANALYSIS: SEED-AVERAGED PER-CASE PAIRED TESTS (TWO-SIDED WILCOXON; N=18 PANCREAS, N=20 LA): MEAN DIFFERENCE, RAW p, AND HOLM-ADJUSTED p_H. HD95 IS THE PRIMARY ENDPOINT AND DICE SECONDARY; FOR EACH COMPARATOR AND METRIC, HOLM CORRECTION IS APPLIED ACROSS THE THREE SETTINGS.

	HD95 (primary)			**Dice** (secondary)		
	Δ	p	p_H	Δ	p	p_H
PEM − BCP						
Pancreas 20%	-1.94	$< .001$	$< .001$	$+1.02$	.021	.042
LA 5%	-6.12	.030	.059	$+0.66$	.090	.090
LA 10%	-3.02	.033	.059	$+0.62$	.014	.041
PEM − PL-FT						
Pancreas 20%	-1.81	.001	.003	$+0.73$	.060	.180
LA 5%	-2.33	.701	.701	-0.26	.927	1.00
LA 10%	$+0.50$	.202	.405	-0.07	.648	1.00
PEM − SC						
Pancreas 20%	-1.75	$< .001$	.003	$+0.87$	.048	.145
LA 5%	-3.99	.077	.153	$+0.24$	.261	.261
LA 10%	-1.17	.097	.153	$+0.44$	.058	.145

VII. DISCUSSION AND CONCLUSION

The BCP checkpoints retain residual entropy concentrated among low-confidence voxels; ρ_f quantifies that concentration, while $\bar{d}$ and the entropy maps establish boundary localization. On these checkpoints, under a minute of label-free fine-tuning reduces mean HD95 by 25 to 44% and raises Dice by $+0.6$ to $+1.0$. Both predicted failure regimes appear: diffuse entropy, visible from one forward pass ($\rho_f = 0.55$, where the Pancreas-selected setting loses 58 Dice), and saturated accuracy on Left Atrium, where PEM's margin over pseudo-labelling vanishes.

Practical guidance. If $\rho_f(0.99)$, from one label-free forward pass, clears $\rho^\star$ and the base Dice leaves headroom, fine-tune at $\tau_m = 0.90$, the only threshold that degraded no SSL checkpoint here, at 5×10^{-6} for batch-normalized checkpoints (destroyed at 5×10^{-5}) and up to 5×10^{-5} for instance-normalized ones.

Three limitations remain: the significance tests are exploratory (Sec. VI), and even there separate PEM from pseudo-label fine-tuning only in Pancreas HD95; $\rho^\star$ rests on four checkpoints; and 18 to 20 test volumes limit power.

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